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FoV-Net: Rotation-Invariant CAD B-rep Learning via Field-of-View Ray Casting

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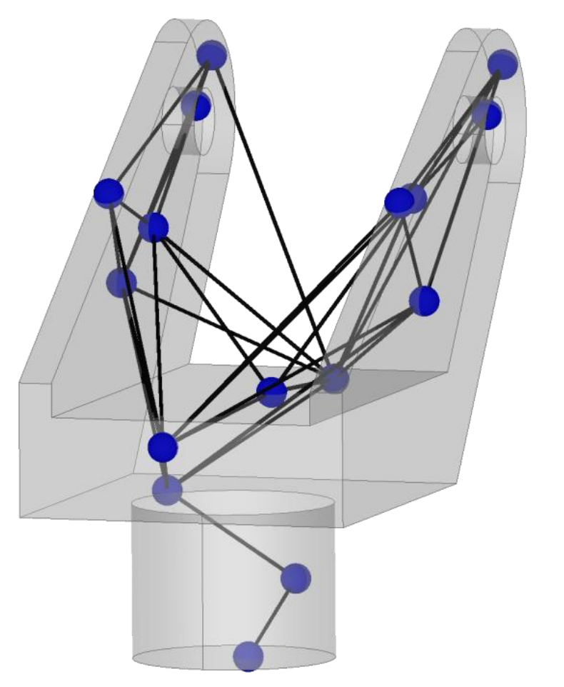
ABSTRACT

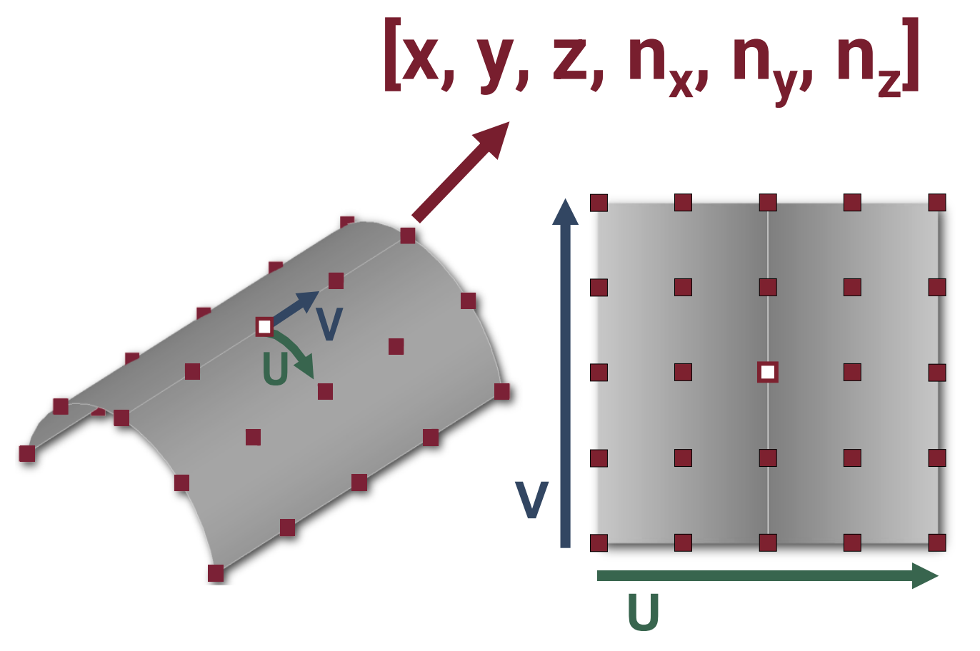
We quantify the rotation sensitivity of current deep learning methods for CAD Boundary representations (B-reps) and introduce FoV-Net as the **first rotation-invariant framework for B-rep learning**. FoV-Net combines Local Reference Frame (LRF) UV grids with Field-of-View (FoV) ray-casting descriptors to capture local geometry and global structure independent of global pose. Features are encoded with lightweight CNNs and aggregated over the B-rep graph using a graph attention network. Experiments show **strong results, robustness to SO(3) rotations** and **improved data efficiency** over existing methods.

MOTIVATION

B-reps are commonly represented as **face-adjacency graphs (a)**, where nodes correspond to surfaces and edges encode topological adjacency. Existing learning methods construct node features for each face by **sampling surface coordinates and normal vectors from each surface's UV parameterization (b)**. Because these features are expressed in the global coordinate frame, the representation is **inherently pose-dependent and tied to the model's orientation**. If the part was designed upside-down, the position and orientation of each surface would change. Consequently, arbitrary SO(3) rotations of an otherwise identical CAD model produce inconsistent inputs, despite unchanged geometry, which we show leads to **severe performance degradation**. FoV-Net instead encodes geometric quantities in local reference frames, and probes surrounding surfaces with rays for structural information.



(a) Face adjacency graph



(b) UV surface sampling

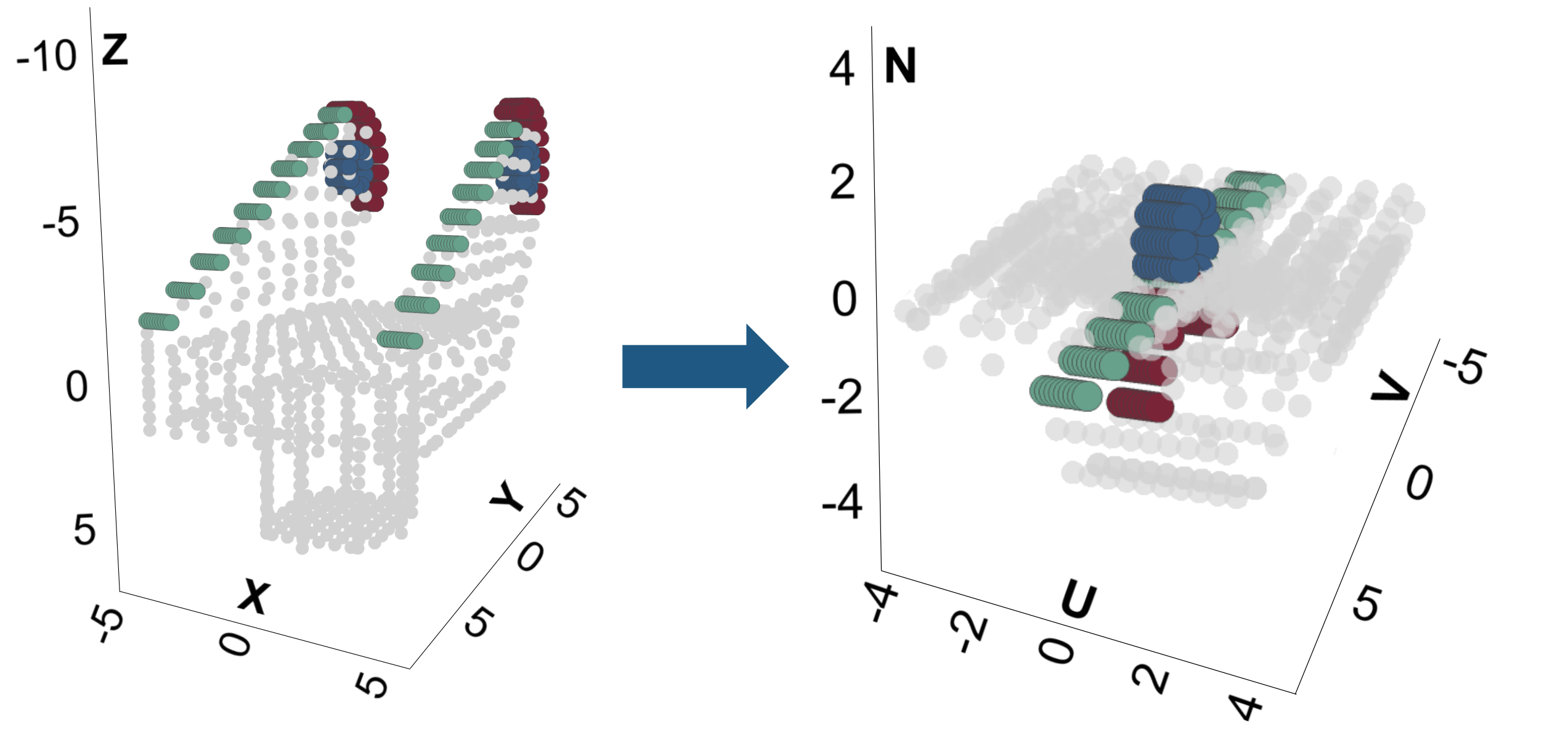
METHODOLOGY

LOCAL-REFERENCE-FRAMES (LRF)

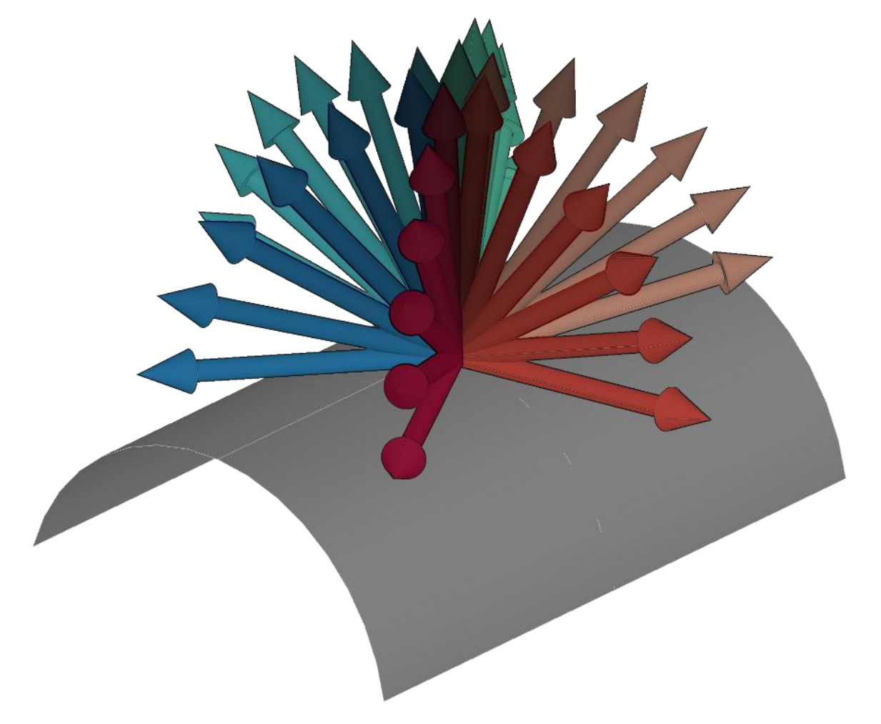
The Local Reference Frame (LRF) UV-grid encodes each CAD face in a rotation-invariant, face-centric coordinate system. For every face, we construct an orthonormal LRF at its center using the surface normal (N) and two tangent directions (U and V) derived from the UV parameterization. This frame moves with the geometry, removing dependence on global pose. Like in (b), we sample the UV domain and obtain coordinates and normal vectors, but now relative to those of the surface center. **Identical surfaces produce identical LRF-UV grids regardless of where they are in 3D space, denoted by colours in (c).**

FIELD-OF-VIEW (FOV) RAY CASTING

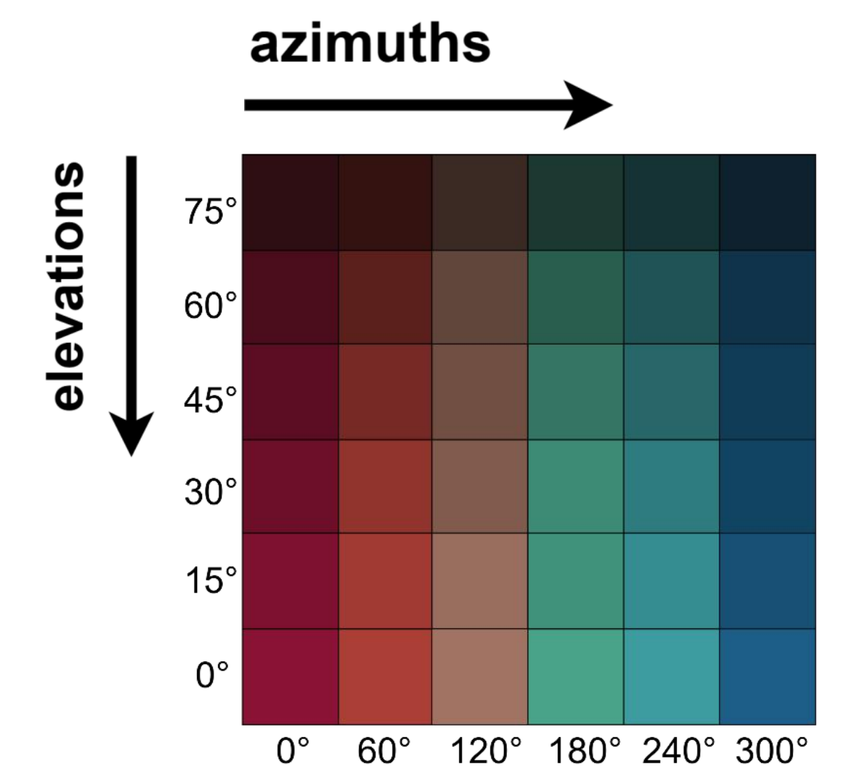
While the LRF removes dependence on global pose, it also loses relative orientations between faces. The Field-of-View (FoV) descriptor restores this structural context using ray casting. Rays are emitted from the face center over a hemisphere aligned with the LRF and discretized into elevation and azimuth bins. For each direction, we store the first intersection with the B-rep, including a hit flag, distance, and incidence angle. This produces a structured grid capturing both outward (external) and inward (internal) visibility. Because all rays are defined in the local frame, **the descriptor remains rotation-invariant while encoding the surrounding 3D structure.**



(c) From UV to LRF-UV grids

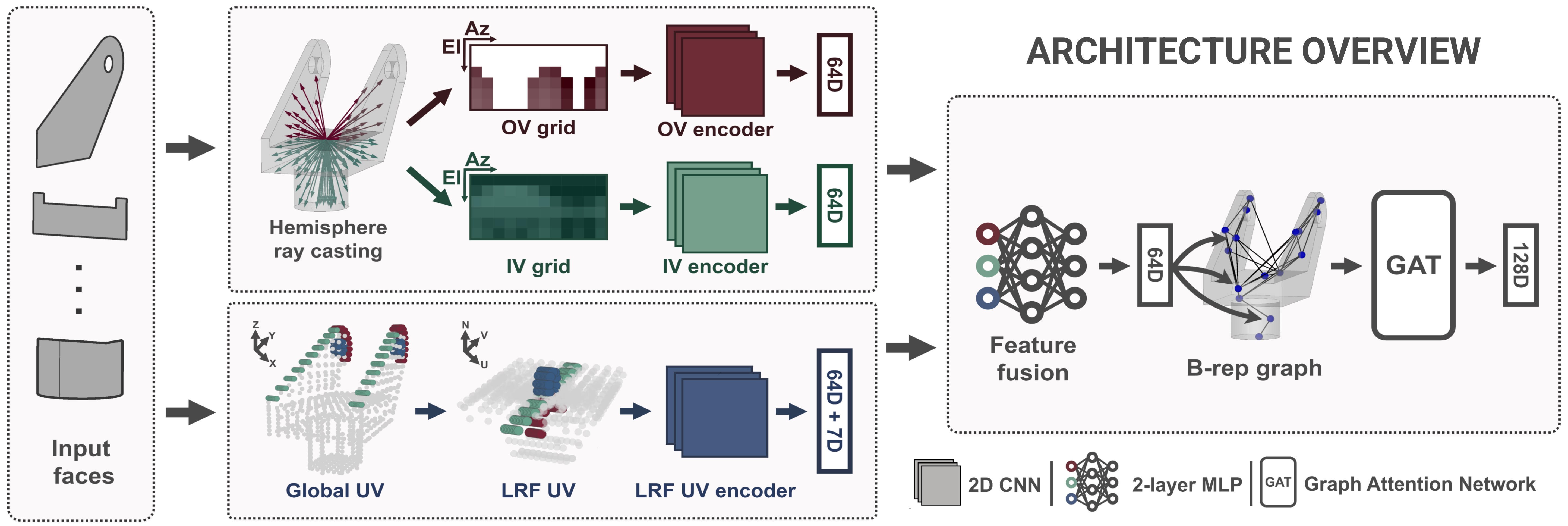


(d) Hemisphere ray casting



(e) Mapping rays to a 2D grid

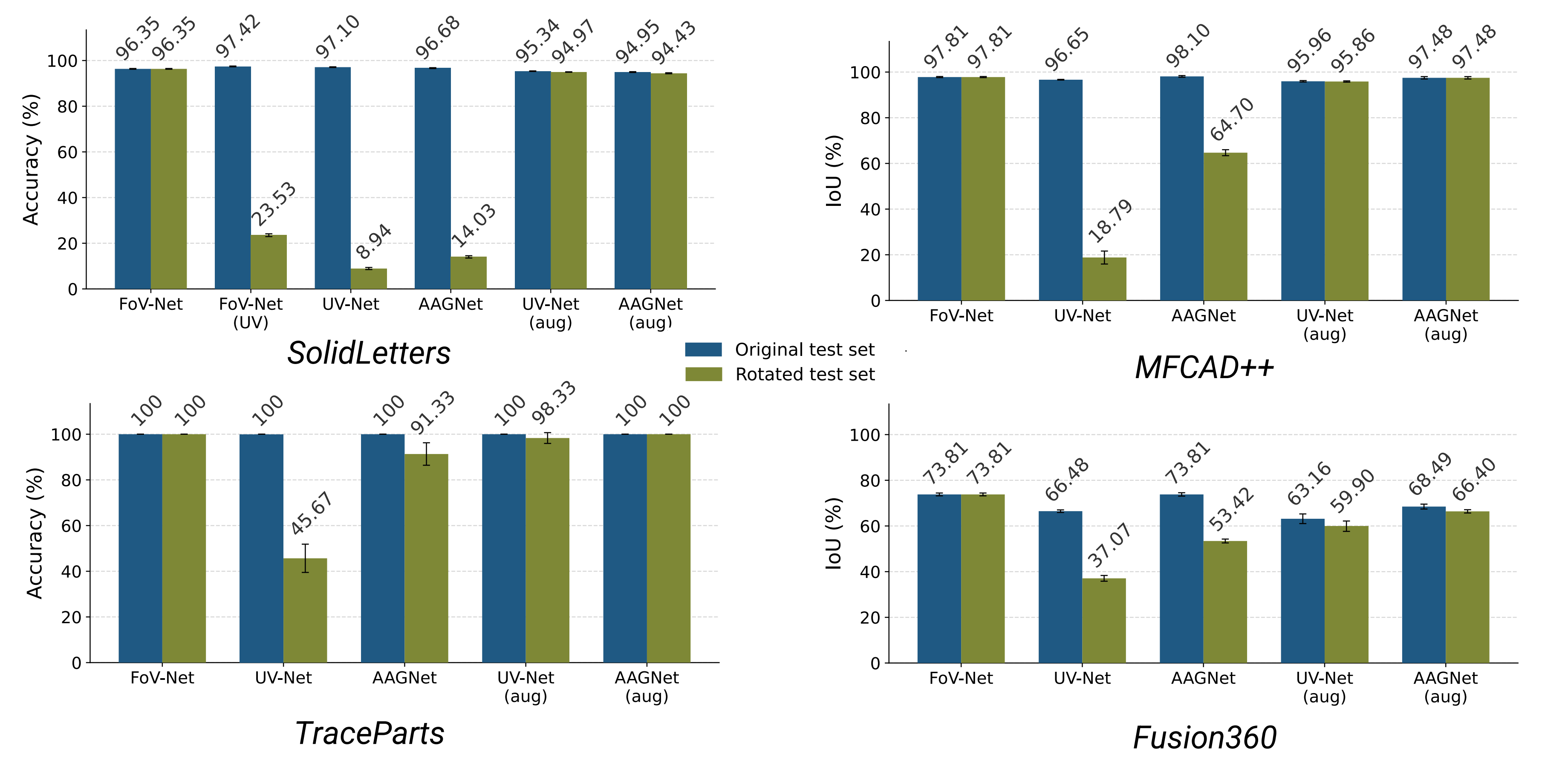
ARCHITECTURE OVERVIEW



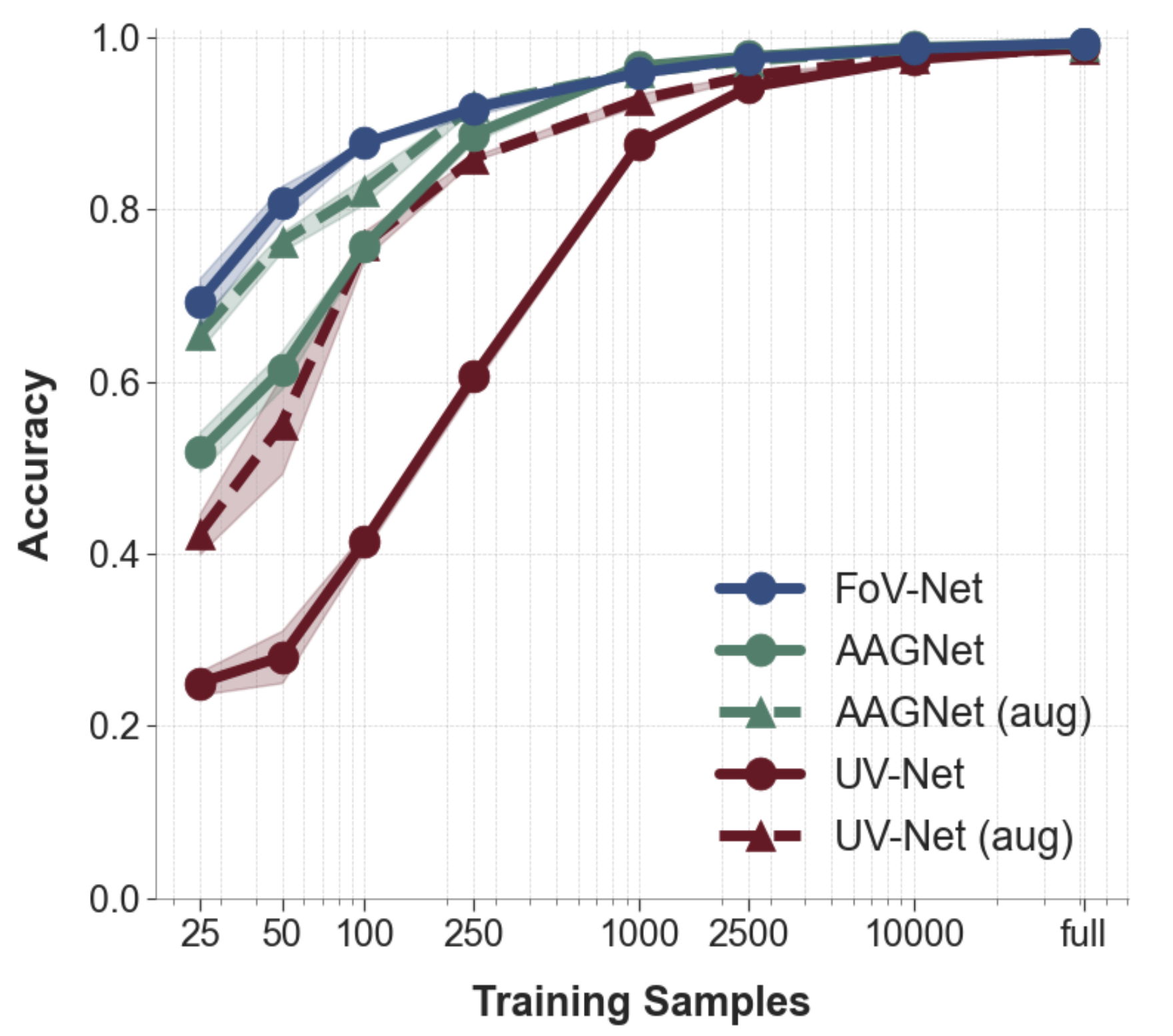
RESULTS

SHAPE CLASSIFICATION

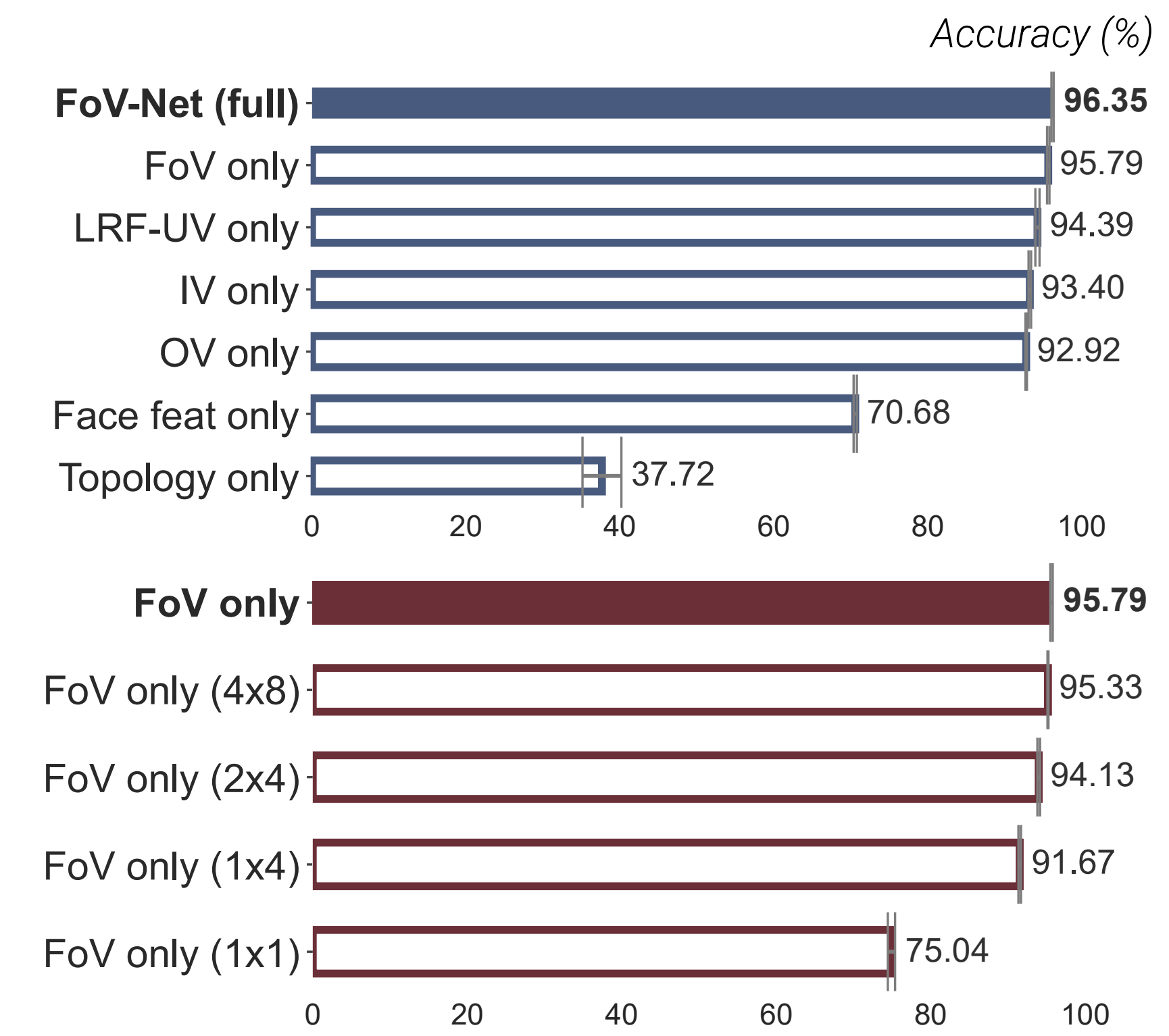
SURFACE SEGMENTATION



DATA EFFICIENCY



ABLATION STUDIES



CONCLUSION

Current B-rep learning methods rely on absolute coordinates and orientations, causing **severe performance degradation under arbitrary rotations.**

Rotation augmentation improves robustness, but reduces performance.

FoV-Net shows **strong, rotation-invariant performance and superior data efficiency**, an important advantage for industrial CAD applications where labeled data is often limited.

CONTACT